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Deep convection in the glacial Pacific can hinder oceanic carbon sequestration: An ocean-biogeochemical model study

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Abstract Glacial atmospheric pCO_2 declines significantly relative to interglacials, with the ocean serving as the primary sink for atmospheric carbon loss. Geological records indicate an increase in carbon storage in the glacial North Pacific, making it an important contributor to the pCO_2 decline. The climate-carbon cycle dynamics driving the increase in deep North Pacific carbon storage remain highly controversial; compared to the abundant data available for the North Atlantic, the research foundation is particularly weak. Based on paleoceanographic carbon cycle proxies from North Pacific sites, previous studies have proposed the "deep convection hypothesis", suggesting that deep convection may have existed in the glacial North Pacific, enhancing surface nutrient utilization efficiency and thereby increasing deep carbon storage. However, existing research lacks a quantitative understanding of the formation conditions for glacial North Pacific deep convection, its carbon cycle dynamics, and its impact on atmospheric pCO_2. To address the relationship between glacial North Pacific deep convection and air-sea carbon balance, here we employed a state-of-the-art ocean carbon cycle model to conduct a series of anti-hosing experiments in the North Pacific. We show that, with Southern Ocean processes held constant, the presence of deep convection in the glacial North Pacific would disrupt the North Pacific halocline structure, induce diapycnal mixing, and release carbon from the deep Pacific reservoir, thereby hindering oceanic carbon storage and ultimately attenuating the glacial decline in atmospheric pCO_2—inconsistent with predictions of the "deep convection hypothesis". Conversely, with the presence of intermediate-depth convection in the glacial North Pacific, redistribution of surface alkalinity and dissolved inorganic carbon concentrations can reduce the pCO_2 difference between the sea surface and atmosphere, making the North Pacific a carbon sink, aligning with geological records. Further comparisons between numerical simulations and records highlight the complexity of glacial North Pacific carbon cycle processes, necessitating improvements to North Pacific salinity flux schemes in future numerical simulations, while incorporating paleoenvironmental records and taking into account the location, intensity, and depth of possible convection processes.

Keywords Carbon cycle, Numerical modeling, Paleoclimatology, Glacial cycles

1. Introduction

Glacial-interglacial cycles represent the most prominent feature of Earth's surface system over the late Cenozoic. In the first half of the twentieth century, the Serbian scientist Milankovitch proposed that summer solar radiation at 65°N latitude is modulated by Earth's orbital parameters—eccentricity, obliquity, and precession—and that its periodic variations control surface climate's glacial-interglacial cycles (Milankovitch, 1941). In the second half of the twentieth century, John Imbrie, Andre Berger, and others further suggested that internal feedback processes within the surface system in response to solar radiation forcing play a key role in cold-warm transitions during glacial-interglacial cycles (Berger, 1988; Imbrie et al., 1992, 1993). Among these, atmospheric pCO_2, as a key greenhouse gas, directly influences surface energy balance and serves as an important internal feedback mechanism.

Tiny air bubbles preserved in high-latitude polar ice caps from both hemispheres reflect the atmospheric chemistry during the ice formation process, serving as an important archive for reconstructing past atmospheric pCO_2 concentrations. Antarctic ice cores preserve the longest continuous record of atmospheric chemical composition to date, advancing accurate understanding of atmospheric CO_2 concentrations back to approximately 800,000 years ago (Lüthi et al., 2008). Ice core records show periodic fluctuations in atmospheric pCO_2 during glacial-interglacial cycles, with amplitudes reaching 100 ppm (1 ppm = one part per million by volume) (Lüthi et al., 2008). Relative to interglacial levels of about 260–280 ppm atmospheric pCO_2, this means that during the glacial cooling, the atmospheric carbon reservoir lost approximately one-third of its total inventory, equivalent to ~200 Gt (1 Gt = 10^{15} g). The driving mechanisms for the CO_2 decline during glacial cooling have been one of the most pressing scientific questions in climate change and the broader Earth sciences over the past three decades (Sigman and Boyle, 2000; Sigman et al., 2010).

On glacial-interglacial timescales, atmospheric pCO_2 is controlled by multiple factors, but ultimately determined by the carbon budget among the three major reservoirs: land, ocean, and atmosphere (Kump and Arthur, 1999). From a mass balance perspective, the atmospheric loss during glacials must be absorbed by the terrestrial carbon reservoir (including terrestrial

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biomass, soil carbon pools, etc.) and/or the oceanic carbon reservoir. Although controversies persist, it is generally accepted that the terrestrial carbon reservoir lost approximately 300 to 1200 GtC during glacial conditions, as exemplified by the Last Glacial Maximum (LGM, ~21,000 years ago) (Peterson et al., 2014). This implies that the ocean was the primary sink for glacial atmospheric CO_2 loss.

The global ocean is the largest carbon reservoir that directly exchanges with the atmosphere (Pierre et al., 2023). Modern data indicate that the total oceanic carbon inventory is approximately 38,000 GtC, which is about 50–60 times that of the atmosphere. Complex carbon cycle processes operate between the ocean and atmosphere, controlling air-sea carbon exchange through physical and biogeochemical processes, determining the carbon partitioning between the ocean and the atmosphere, and ultimately affecting atmospheric CO_2 concentrations (Hülse et al., 2017). Notably, due to the vast magnitude difference in carbon inventories between the ocean and atmosphere, perturbations in the oceanic inorganic carbon inventory can be amplified in the atmospheric carbon reservoir, with amplification factors reaching 7 (global average) to 14 (Southern Ocean) (DeVries, 2022). Thus, minor disturbances in the oceanic carbon reservoir can have substantial impacts on atmospheric CO_2 concentrations.

Although increases in the glacial oceanic carbon inventory are governed by both physical conditions and biogeochemical processes, it is generally believed that changes in ocean physical conditions dictate secondary biogeochemical responses (Völker and Köhler, 2013), which together affect the inventory of the oceanic carbon reservoir. At global ocean scales, the transformation of thermohaline circulation from modern to LGM conditions had a decisive influence. Under modern conditions, the ocean thermohaline circulation is mainly controlled by the sinking branch formed in the North Atlantic and the upwelling branch in the Southern Ocean (Marshall and Speer, 2012; Talley, 2013), corresponding to the deep water masses formed in the respective regions, namely North Atlantic Deep Water (NADW) and Antarctic Bottom Water (AABW). The thermohaline circulation begins with the formation of NADW in the Atlantic, flows through the Southern Ocean, participates in the formation of AABW, and ultimately reaches the deep North Pacific. No deep water forms in the modern North Pacific (Rae et al., 2020). Paleoceanographic evidence—from water mass tracers (elements/isotopes) to ventilation proxies—indicates that glacial NADW shoaled overall, termed Glacial North Atlantic Intermediate Water (GNAIW) (Curry and Oppo, 2005; Lippold et al., 2012). With GNAIW volume contraction, AABW volume expanded relatively. The oceanic carbon reservoir, primarily composed of dissolved inorganic carbon (DIC), migrates along thermohaline pathways. Glacial AABW expansion coincided with the overall slowdown of Atlantic basin circulation. This "circulation stagnation" concept underpins mainstream theories for glacial Atlantic carbon reservoir expansion (Hain et al., 2010).

Relative to the Atlantic, glacial Pacific carbon cycle dynamics exhibit notable uncertainties (Du and Tian, 2025). Under modern conditions, the Pacific is the largest marine DIC reservoir, accounting for more than 50% of the total ocean inventory. Pacific carbon reservoir formation is directly tied to global thermohaline circulation pathways and North Pacific regional oceanographic characteristics. Along the thermohaline circulation pathway, organic matters are continuously oxidized and degraded to DIC, accumulating into the respired carbon inventory (Sigman et al., 2021), resulting in the highest DIC concentrations at the circulation terminus in the North Pacific. More importantly, the surface salinity in the North Pacific is low, with a well-developed halocline, reducing carbon exchange between surface and deep waters, thereby causing more DIC to accumulate further in the deep North Pacific. Geological evidence shows reduced deep Pacific dissolved oxygen during glacials (Jacobel et al., 2017), implying expanded Pacific carbon storage and thus contributions to glacial atmospheric pCO_2 declines.

Mechanisms for glacial Pacific carbon inventory increases represent a current hotspot and research gap in carbon cycle studies. Based on radiocarbon isotopes and other paleoceanographic evidence (Skinner et al., 2015; Du et al., 2018), prior mainstream views posited slower or stagnant glacial Pacific circulation, allowing extended respired DIC accumulation in the deep Pacific and thus enhancing storage. This hypothesis essentially posits similar carbon storage mechanisms in the glacial Pacific and Atlantic, namely, slowing the thermohaline upwelling branch (i.e., the AABW cell) via cooling or Southern Hemisphere westerly shifts (Watson et al., 2015). However, Nd isotope data and our prior simulations suggest (Hu and Piotrowski, 2018; Du and Tian, 2025) that glacial deep Pacific circulation, driven by marked Southern Ocean cooling, may have been relatively vigorous—i.e., deep water mass residence times may not have increased but decreased, disfavoring deep carbon storage. Conversely, vigorous circulation could shorten air-sea carbon exchange times for upwelled deep carbon-rich waters in the Southern Ocean. Since Southern Ocean surface pCO_2 exceeds atmospheric levels, reduced exchange time retains CO_2 in the ocean that would otherwise outgas, effectively boosting storage; this deviation from chemical equilibrium is termed the disequilibrium pump (Khatiwala et al., 2019). Vigorous circulation and sea ice expansion further shorten Southern Ocean surface carbon exchange times, enhancing regional disequilibrium pump efficiency, potentially offsetting reduced accumulation time and ultimately increasing Pacific storage.

However, the greatest uncertainty in glacial North Pacific carbon cycle dynamics centers on changes in the North Pacific halocline and its role in local and global carbon cycles. Due to the special sedimentary conditions in the North Pacific limiting the acquisition of research materials, combined with some numerical simulation results (Menviel et al., 2012, 2014), it is generally thought that the glacial North Pacific still maintained or even strengthened a stable halocline structure, thereby favoring carbon storage in the Pacific reservoir. But a recent study has questioned this understanding (Rae et al., 2020). The study found that intermediate water carbon isotopes ($\delta^{13}C$) at a number of North Pacific sites are more enriched relative to the late Holocene (Fig. 1A, Fig. 1F). This is consistent with redox indicators (Fig. 1B), radiocarbon isotope ages ($\Delta^{14}C$, Fig. 1C), and other records, indicating improved ventilation in the North Pacific. Meanwhile, sea surface temperature (SST) and sea surface salinity (SSS) both increased (Fig. 1D, Fig. 1E). These findings suggest that intermediate-deep convection may have existed in the glacial North Pacific, also meaning that the surface halocline in the North Pacific may have weakened or disappeared. The study further inferred that halocline weakening may have contributed to the decline in atmospheric pCO_2. In contrast, numerical simulation experiments with an intermedi-

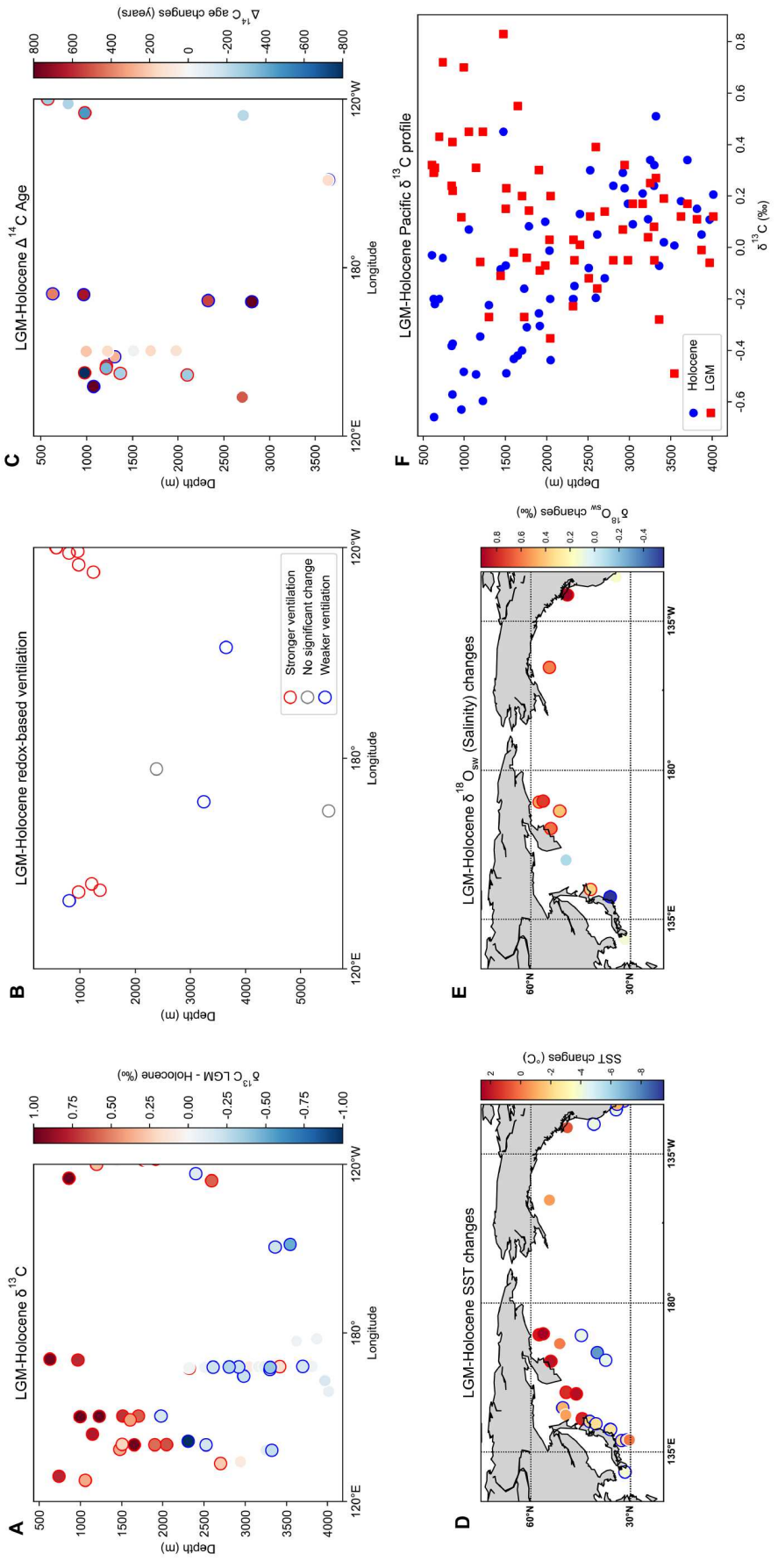


Fig. 1 Overview of glacial North Pacific carbon cycle. (A) $\delta^{13}C$ records north of 20°N in the North Pacific. In the figure, the shading in the circles represents the absolute change from LGM relative to Holocene, and the edge color represents the change trend—red for increase, white for no obvious change, blue for decrease. (B) Ventilation changes indicated by redox indicators north of 20°N in the North Pacific, with red for improved ventilation, gray for no obvious change, blue for worsened. (C) $\Delta^{14}C$ age changes north of 20°N in the North Pacific. In the figure, the shading in the circles represents the absolute change from LGM relative to Holocene, and the edge color represents the change trend—red for increase, white for no obvious change, blue for decrease. (D and E) respectively depict the changes in SST and seawater $\delta^{18}O$ (indicating salinity) from LGM relative to Holocene. (F) data from (A) are plotted as depth profiles, red for LGM, blue for Holocene. The figure shows that upper ocean water masses held enriched $\delta^{13}C$ imprints during LGM. Data are compiled from Rae et al. (2020).

ate-complexity model reveal distinct processes based on modern climate backgrounds (Menviel et al., 2012, 2014): in these experiments, the direct effect of halocline weakening is to simultaneously bring nutrients and DIC to the upper North Pacific. The increase in nutrient supply, especially surface iron, stimulates marine primary productivity, thereby potentially increasing carbon sink effects; however, the carbon consumed by the increase in primary productivity is insufficient to offset the CO_2 brought from the deep, making the nutrient utilization efficiency decrease, suggesting that halocline weakening is unfavorable for atmospheric pCO_2 decline. The opposing understandings between geological records and simulation results regarding the carbon cycle effects of the North Pacific halocline highlight our insufficient understanding of glacial North Pacific carbon cycle dynamics; therefore, conducting coupled ocean-carbon cycle numerical simulations under paleoclimate backgrounds can clarify the complex carbon cycle processes therein and deepen our understanding of the evolution of ocean carbon sinks.

Notably, under paleoclimate backgrounds, numerical simulations of North Pacific thermohaline circulation do not align with geological records. For instance, at the Pliocene-Pleistocene transition, paleoproductivity and sedimentation proxies suggest that the North Pacific may have hosted North Pacific Deep Water (NPDW) formation similar to modern NADW (Haug et al., 1999; Burls et al., 2017; Ford et al., 2022), extending southward to form the Pacific Meridional Overturning Circulation (PMOC) originating from the high-latitude North Pacific. However, in most simulations under Pliocene conditions, models fail to spontaneously simulate PMOC; it is only observed in a few experiments with specific tectonic backgrounds (Tan et al., 2022) or specific SSS perturbations (Fu and Fedorov, 2024). This discrepancy arises from the broad oceanic topography of the North Pacific and abundant net precipitation (Burls et al., 2017), which keeps the North Pacific surface salinity too low to break stratification and form deep convection. The existence of PMOC during glacials remains a debated topic, with even less understanding. In intermediate-complexity Earth system models without flux correction (Buchanan et al., 2016) or coupled climate models (Zhang et al., 2013), the deep Pacific stratification strengthens during the LGM. Therefore, although geological studies (Rae et al., 2020) use multiple proxies to infer the existence of deep convection in the glacial North Pacific, the authors rely on preliminary verification with intermediate-complexity models using salinity flux corrections. In other words, discussions based on general circulation models regarding the carbon cycle effects of glacial PMOC are still lacking in paleoceanographic carbon cycle research.

To address the role of the glacial North Pacific halocline in carbon storage, we use a state-of-the-art ocean-biogeochemical model with which we follow previous North Pacific salinity sensitivity experiments to conduct a series of sensitivity experiments that produce distinct circulation and ventilation patterns in the North Pacific, thereby revealing significant responses in oceanic carbon storage.

2. Methods

2.1 Model description

Here we employ an ocean-biogeochemical model, MITgcm-REcoM2. The physical component is an ocean general circulation model MITgcm (Campin et al., 2020), which includes a thermodynamic-dynamic sea ice module. MITgcm utilizes the subgrid mesoscale eddy parameterization scheme proposed by Gent and McWilliams (Gent et al., 1995), the KPP turbulent mixing scheme proposed by Large (Large et al., 1994), and the bottom boundary scheme proposed by Campin and Goose (Campin and Goosse, 1999). The model topography covers most ocean regions from 80°S to 80°N, thus excluding the Arctic Ocean. The model has a horizontal grid of 180×126 cells, with a longitudinal resolution of 2°. The latitudinal resolution is 2°; yet, to better simulate carbon cycle processes in the Southern Ocean and tropical regions, we enhanced the resolution near the Southern Ocean and equator, with the highest resolution reaching 0.38°. The maximum depth is 5500 m, separated into 30 uneven layers, with thicknesses gradually increasing from 10 m at the surface to a maximum of 500 m in the deep ocean.

The ocean biogeochemical model REcoM2 is a typical NPZD-type model (Nutrient-Phytoplankton-Zooplankton-Detritus) (Hauck et al., 2013). The nutrients in the model include dissolved inorganic nitrogen (DIN), dissolved silicon (DSi), dissolved iron (DFe), etc. The model includes two types of phytoplankton: larger diatoms and relatively smaller phytoplankton (some of which are calcifiers). Calcifying phytoplankton, while performing photosynthesis, use DIC from surrounding waters to generate calcium carbonate, simultaneously releasing CO_2. REcoM2 includes one type of zooplankton group, which consumes organic carbon and nitrogen produced by phytoplankton, generating corresponding inorganic products. Detritus includes components such as particulate nitrogen, particulate carbon, insoluble iron, and calcium carbonate; the sinking velocity of these particulate matters in the ocean increases with depth.

The seawater carbonate system follows the protocol of the Ocean Carbon-cycle Model Intercomparison Project (OCMIP; http://www.ipsl.jussieu.fr/OCMIP/). The air-sea CO_2 exchange follows Wanninkhof (1992); the actual flux is inversely proportional to the sea ice concentration (SIC) in the grid cell.

Compared to other biogeochemical models, REcoM2 has the following advantages: first, primary production allows for flexible nutrient ratios, enabling phytoplankton to adapt to different environmental conditions; second, it explicitly simulates iron cycling, reproducing iron limitation in the Southern Ocean, allowing for simulation of glacial iron fertilization effects; third, the sea ice module improves the simulation of carbon dynamics in the high-latitude Southern Ocean.

The MITgcm-REcoM2 has been used to investigate the modern and paleo ocean carbon cycles, including decadal scales (Hauck and Völker, 2015), glacial-interglacial cycles (Du et al., 2022; Du and Tian, 2025), and tectonic scales (Du et al., 2025).

2.2 Experimental design and initialization

This paper focuses on the mechanisms of glacial Pacific carbon storage, emphasizing the impact of North Pacific halocline changes on physical and biogeochemical processes. To this end, we conducted three experiments with MITgcm-REcoM2: one control experiment (LGM_ctl) and two sensitivity experiments (LGM_pac_005, LGM_pac_010).

The control experiment LGM_ctl (referred to as the "Glacial Experiment") uses published LGM boundary conditions (Zhang et

al., 2013), representing a typical glacial carbon cycle state. Compared to modern conditions, topography in LGM_ctl is adjusted for a 116 m sea-level drop, with minor tweaks to ensure numerical stability. Initialization is divided into physical and biogeochemical parts. The physical initialization uses temperature and salinity from WOA09, interpolated to LGM conditions (Antonov et al., 2010; Garcia et al., 2010; Locarnini et al., 2010). The biogeochemical initialization uses DIN and DSi from WOA09, alkalinity and DIC from GLODAP (Olsen et al., 2016), and iron from Aumont et al. (2015), with volume adjustment for LGM.

The sensitivity experiments LGM_pac_005 (referred to as the "Shallow Convection Experiment") and LGM_pac_010 (referred to as the "Deep Convection Experiment") apply negative freshwater fluxes to LGM_ctl in the North Pacific between 45°–50°N and 150°–230°E, at −0.05 Sv and −0.1 Sv (1 Sv = 10^6 m^3 s^{-1}), artificially raising surface salinity to weaken or destroy the glacial North Pacific halocline.

The experiments were first run with a physics-only setup for 5000 years to reach equilibrium in the deep oceans. Biogeochemistry was then coupled for another 2000 years until air-sea carbon exchange is balanced (with the change in atmospheric pCO_2 over the last 100 years less than 0.5 ppm). The atmospheric forcings are derived from published coupled LGM simulations (Zhang et al., 2013).

3. Results and discussion

3.1 Overview of the physics and biogeochemistry in the Glacial Experiment

We take the Glacial Experiment under LGM boundary conditions (LGM_ctl) as a typical representative of the glacial carbon cycle and the benchmark for sensitivity experiments. We have detailed the model calibration in previous work (Du et al., 2022; Du and Tian, 2025); here, we provide a brief overview of the physics and carbon cycle state relevant to the sensitivity experiments.

The mean SST in the Glacial Experiment is 15.1 °C, 2.9 °C lower than the simulated interglacial conditions. Figs. 2A and 2C show the simulated glacial SSS and SST patterns. Compared to the interglacial conditions, the glacial SST decreases globally, with the most substantial cooling in high-latitude oceans, particularly in the North Atlantic and Southern Ocean. These changes are consistent with proxy reconstructions (Tierney et al., 2020). Dramatic cooling at high latitudes in both hemispheres promoted sea ice expansion. In the Glacial Experiment, the maximum winter sea ice coverage in the Southern Ocean reached 31.1×10^6 km^2, approximately twice that of the Interglacial Experiment; even in Southern Hemisphere summer, Southern Ocean sea ice coverage reached 17.8×10^6 km^2. This substantial expansion of sea ice area contributed to changes in thermohaline circulation in the Glacial Experiment (Du and Tian, 2025). In the Glacial Experiment, the simulated AMOC (Fig. 2F) reached a depth of approximately 2300 m with a strength of about 14 Sv. Both its depth and strength were reduced compared to the Interglacial Experiment. In the Glacial Experiment, there was no deep convection or deep water formation in the North Pacific (Fig. 2G).

In the Glacial Experiment, the model simulates an atmospheric pCO_2 of 223 ppm (Table 1), a decline of 56 ppm compared to the interglacial conditions, which is broadly consistent with existing simulations that do not incorporate additional alkalinity (ALK) boosting; this slightly underestimates the glacial-interglacial atmospheric pCO_2 change. The Glacial Experiment shows an increase in ocean carbon storage of approximately 112 GtC compared to the interglacial conditions, with about two-thirds stored in the ocean below 2000 m. Relative to the interglacial conditions, the global export production in the Glacial Experiment is 4.6 GtC yr^{-1}, a decline of about 1.4 GtC yr^{-1} compared to the interglacial conditions; however, due to enhanced glacial dust iron input (Yamamoto et al., 2019), the decline in export production in the Southern Ocean and North Pacific is smaller than global levels, making these two regions sensitive to iron fertilization in the Glacial Experiment (Du et al., 2022).

3.2 Basin-scale oceanic response driven by salinity forcing

Relative to the Glacial Experiment, the oceanic physical processes showed obvious responses to anti-hosing forcing in the North Pacific in the sensitivity experiments. The direct result of freshwater extraction is an increase in sea surface salinity (SSS, Fig. 3) in the North Pacific, with the core area located between 30°N and 60°N in the North Pacific, amplitudes reaching 1.5 psu (LGM_pac_005) and 2.7 psu (LGM_pac_010). It is noteworthy that while the basin salinity in the North Pacific increases, the salinity response in the North Atlantic at the same latitude differs in the two sensitivity experiments. In the Deep Convection Experiment, the North Atlantic salinity decreases by about 0.4 psu; in the Shallow Convection Experiment, the North Atlantic surface salinity change is not significant. The basin-scale mirrored salinity differences in the deep convection experiment are basically consistent with previous work (Saenko et al., 2004; Rae et al., 2020). In the sensitivity experiments, the Southern Ocean surface salinity shows no significant change.

The changes in SSS lead to increased surface density in the North Pacific, further resulting in deepening of the annual mean mixed layer depth (MLD, Fig. 4). In the Shallow Convection Experiment and Deep Convection Experiment, the North Pacific MLD significantly deepens by 300–500 m within the freshwater extraction area (45°N–50°N). Corresponding to the salinity response, the North Atlantic MLD shows a decreasing trend in the Deep Convection Experiment by ~200 m. Unlike the salinity feedback, the Southern Ocean MLD also shows basin-scale differences in the Deep Convection Experiment. South of 60°S, the MLD in the Ross Sea significantly shallows. The Weddell Sea MLD shows latitudinal differences: MLD deepens south of 75°S, while it shallows between 60° and 75°S.

Changes in MLD are closely related to deep water formation and thermohaline circulation processes. In the Glacial Experiment, there is no deep water formation in the North Pacific, nor is there an obvious Pacific Meridional Overturning Circulation (PMOC) (Fig. 2G). Water masses from the high-latitude Southern Ocean enter the North Pacific and then upwell, returning to the surface Southern Ocean. In the Shallow Convection Experiment (Fig. 5A), the convection in the North Pacific strengthens, forming a relatively weak PMOC with a depth of about 1000 m and a strength (north of 40°N) of about 7 Sv; whereas in the Deep Convection Experiment, the PMOC significantly strengthens to ~18 Sv and deepens to 3000 m. Notably, although the artificial salinity flux changes were only applied to the surface North Pacific, the deep Pacific circulation significantly intensifies.

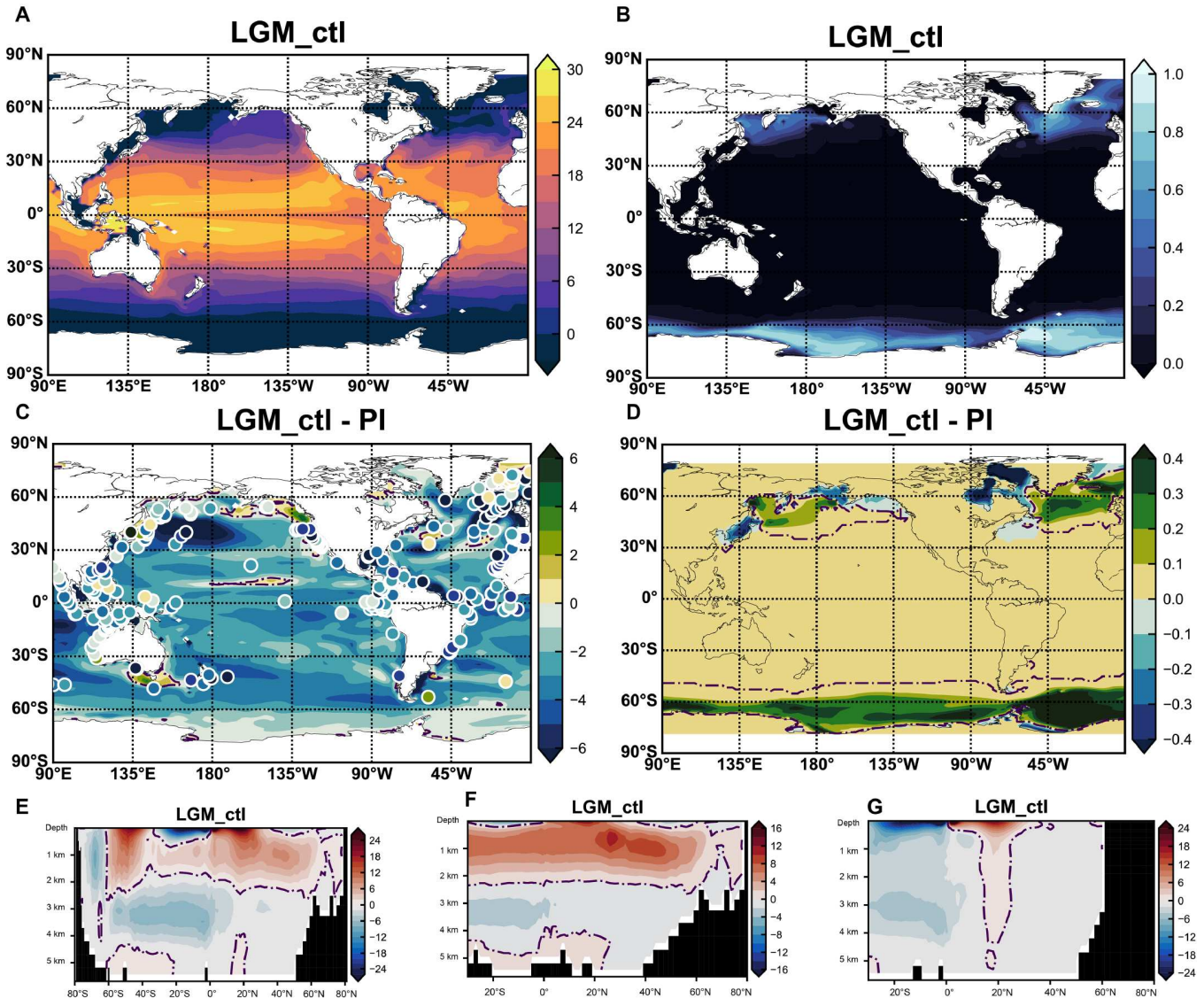


Fig. 2 Outputs of physical fields from the Glacial Experiment. (A) Sea surface temperature (units: °C), (B) Sea ice concentration, (C) SST differences (units: °C) in the Glacial Experiment relative to the Interglacial Experiment; the dots in the figure represent the differences in SST from LGM relative to the late Holocene in a recent LGM data assimilation work (Tierney et al., 2020), (D) Sea ice changes in the Glacial Experiment relative to the Interglacial Experiment, (E) to (G) are the global meridional overturning circulation, Atlantic meridional overturning circulation, and Pacific meridional overturning circulation in the Glacial Experiment, respectively, with all units in Sv.

Table 1 Simulated carbon cycle response in the experiments[a)]

Exp name	LGM_ctl	LGM_pac_005	LGM_pac_010
Fresh-water extraction (Sv)	0	0.05	0.1
Atm. pCO_2 (ppm)	223	+9	+18
Global inventory	36290	−14	−33
Global_b1k[b)] inventory	27922	−112	−272
Pacific inventory	19523	−68	−270
Pacific_b1k inventory	15157	−152	−442
Atlantic	8907	+14	+101
Atlantic_b1k	6813	+10	+75
North Pacific EP (GtC yr^{-1})[c)]	1.34	+0.1	+0.5

a) The (LGM_ctl The column 'LGM_ctl' shows the simulated values in the Glacial Experiment, while in 'LGM_pac_005' and 'LGM_pac_010', the anomalies between the respective experiment and the Glacial Experiment are shown. Unit of carbon inventory change: GtC. b) _b1k refers to 'below 1 kilometer'. c) EP refers to the export production.

The Atlantic Meridional Overturning Circulation (AMOC) (Fig. 5B) shows a mirrored change with PMOC, namely, the circulation throughout the entire water column weakens. The Southern Ocean Meridional Overturning Circulation (SOMOC) and the Global Meridional Overturning Circulation (GMOC) are simultaneously controlled by Pacific and Atlantic processes (Fig. 5C). Overall, SOMOC slightly enhances in both experiments. In the Deep Convection Experiment, consistent with the modeled MLD, the circulation changes near the Antarctic are not significant or even slightly weakened, but the circulation in the 55°S–65°S significantly enhances.

SST changes (Fig. 6A) are closely related to the thermohaline circulation mode. In both convection experiments, due to the formation of PMOC, the cross-equatorial northward heat transport in the Pacific increases, leading to warming in the North Pacific. In the Deep Convection Experiment, the warming in the North Pacific exceeds 4 °C. The weakening of AMOC, on

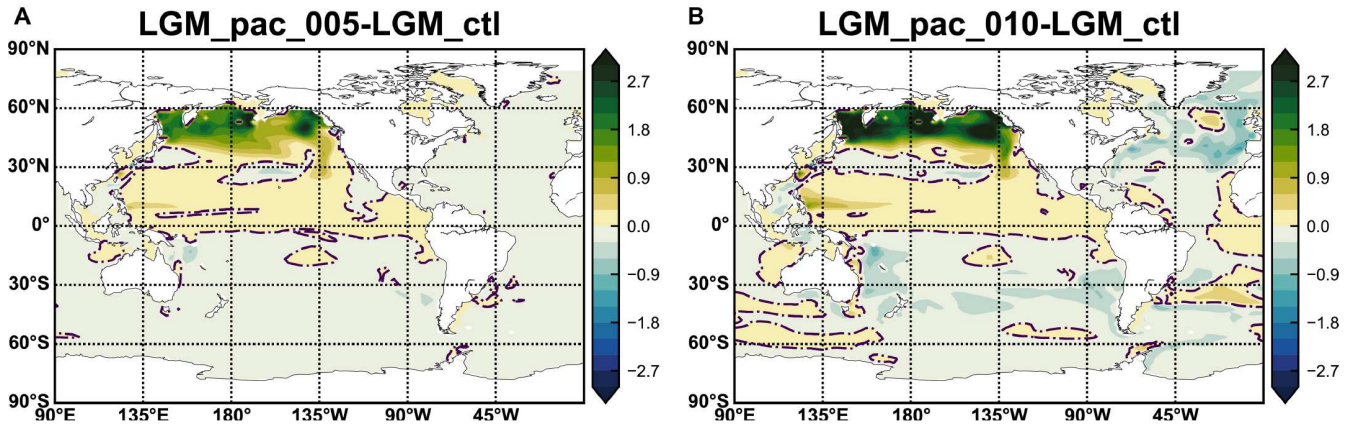


Fig. 3 SSS anomalies in the sensitivity experiments compared to the Glacial Experiment (units: psu).

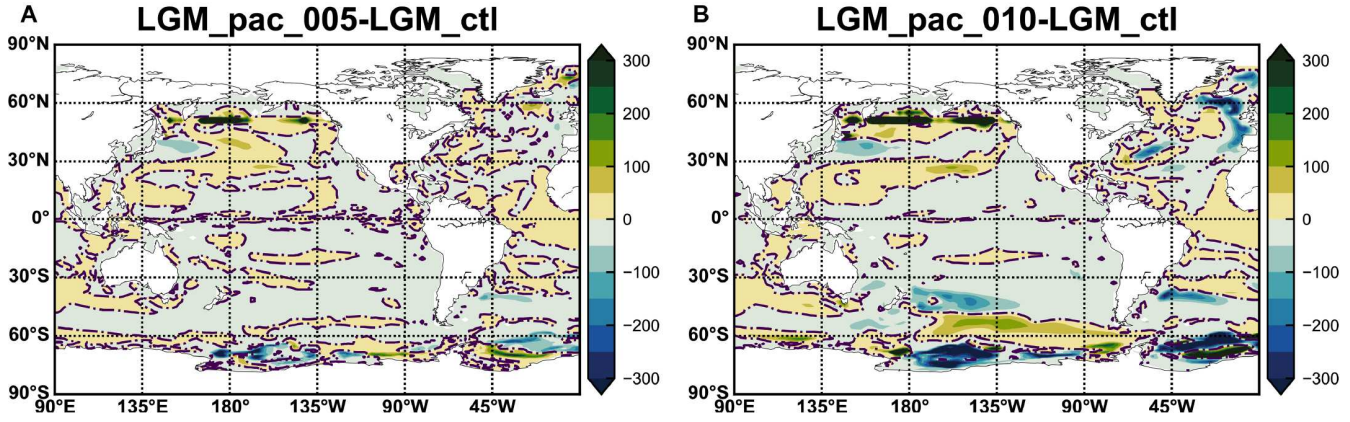


Fig. 4 Annual mean mixed layer depth (MLD) anomalies in the sensitivity experiments compared to the Glacial Experiment (units: m).

the other hand, causes a decline in SST in the North Atlantic. The changes in SST are consistent with the trends in sea ice changes (Fig. 6B). In the sensitivity experiments, the sea ice coverage in the North Pacific, especially the northwestern Pacific, significantly decreases; whereas the sea ice coverage in the Atlantic increases.

Notably, in both sensitivity experiments, the SST in the Southern Ocean increases, and the sea ice coverage decreases, but the deep circulation is enhanced. Since the decrease in sea ice coverage weakens brine rejection, which itself is unfavorable for enhancing the deep circulation, this indicates that more complex dynamical processes may occur in the Southern Ocean. Because Southern Ocean processes are closely related to deep carbon cycle dynamics, we will discuss them together in the section on deep carbon cycle response.

3.3 Basin-scale carbon cycle seesaw

Changes in physical conditions led to responses in the air-sea carbon cycle system (Table 1). In the Glacial Experiment, the modeled atmospheric pCO_2 is 223 ppm, slightly higher than ice core records. Previous analyses have shown (Du et al., 2022) that the deviation is due to rudimentary representation of sediment processes on the ocean carbon cycle in the current version of MITgcm-REcoM2, resulting in a relatively weak carbonate compensation (buffering) effect in the ocean (Hain and Sigman, 2025). Moreover, the model does not consider changes in ocean total alkalinity during glacials (Toggweiler, 2008), which may also be one of the reasons for the deviation. The modeled atmospheric pCO_2 rises with increasing freshwater extraction. In the Shallow Convection Experiment, atmospheric pCO_2 rises by 9 ppm; in the Deep Convection Experiment, atmospheric pCO_2 further rises by 9 ppm relative to the Shallow Convection Experiment, reaching 241 ppm.

The substantial changes in atmospheric pCO_2 are primarily controlled by more drastic alterations in basin-scale carbon storage patterns. The changes in the total ocean reservoir in the Shallow Convection and Deep Convection Experiments are +14 GtC and +33 GtC, respectively, consistent with the variations in the atmospheric carbon reservoir. The Pacific basin shows significant carbon loss in the sensitivity experiments, at −68 GtC and −270 GtC, respectively, whereas the Atlantic basin carbon inventory increases by +14 GtC and +101 GtC. Therefore, although the changes in atmospheric pCO_2 and total ocean carbon storage exhibit near-linear characteristics associated with the North Pacific freshwater forcing, responses in the deep ocean carbon cycle are nonlinear. Meanwhile, the variations in carbon inventory within the basins are not uniform. In the convection experiments, the carbon loss in the Pacific is mainly from the deep ocean below 1000 m, whereas the inventory in the upper 1000 m increases. The Atlantic shows slight differences in the two sensitivity experiments. In the Shallow Convection Experi-

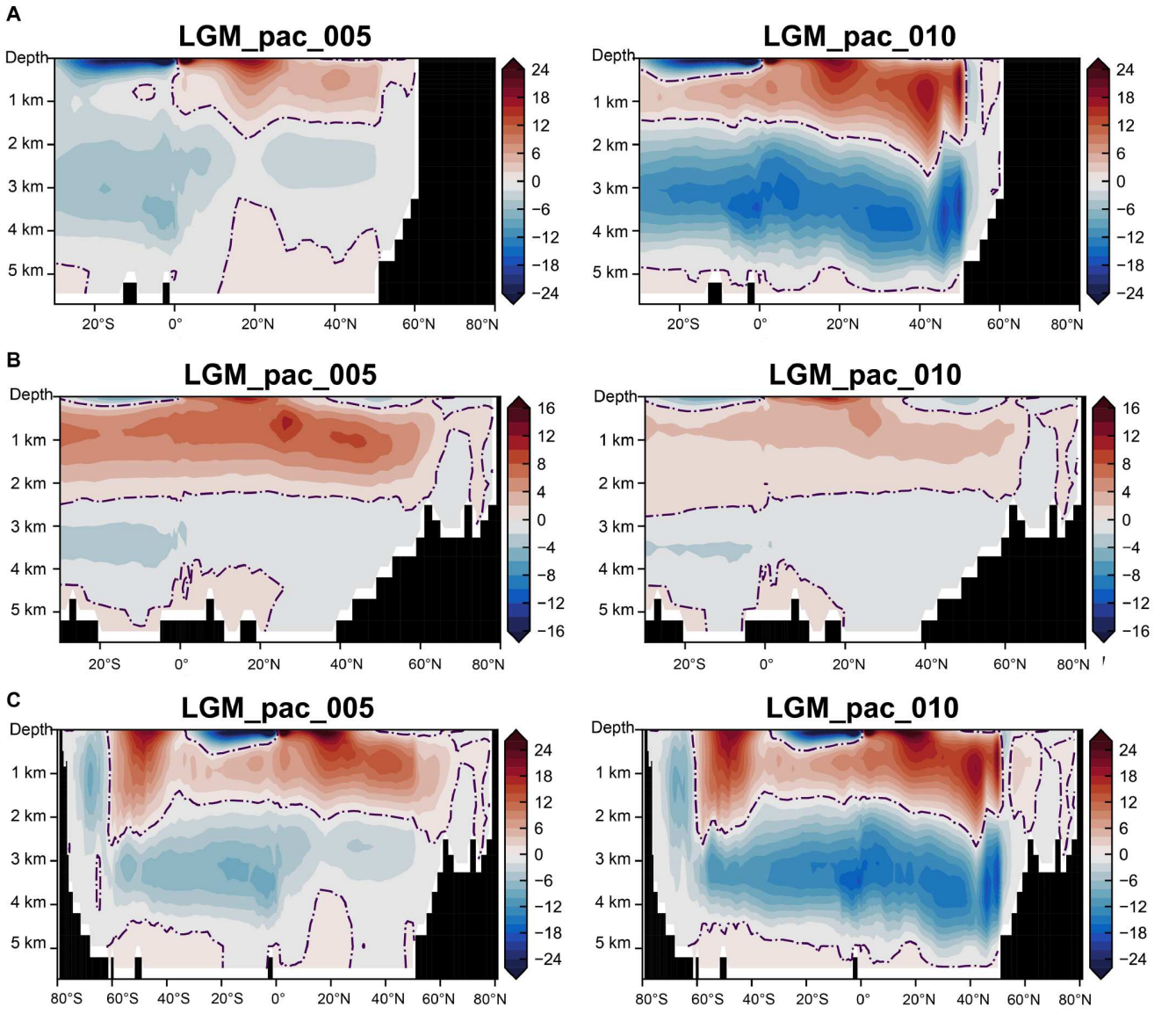


Fig. 5 Meridional overturning circulation in the sensitivity experiments. The left column is the Shallow Convection Experiment, and the right column is the Deep Convection Experiment. (A) Pacific Meridional Overturning Circulation (PMOC); (B) Atlantic Meridional Overturning Circulation (AMOC); (C) Southern Ocean Meridional Overturning Circulation (SOMOC) and Global Meridional Overturning Circulation (GMOC). Units: Sv.

ment, positive DIC anomalies (increased carbon storage) are mainly located in the deep Atlantic and areas controlled by SOMOC; whereas in the Deep Convection Experiment, almost the entire Atlantic carbon storage increases.

The deep carbon inventory in the Northern Hemisphere oceans is controlled by changes in halocline structure (Fig. 7). Taking the Pacific as an example, in both convection experiments, the positive salinity anomalies in the North Pacific penetrate into the ocean interior with the formation of deep convection. From the salinity structure, it can be observed that as the salinity forcing intensifies, saline seawater expands deeper and more southward within the Pacific. The perturbation of the halocline structure weakens the stratification in the Pacific. In the model, we applied a vertical mixing scheme determined by stratification (the KPP scheme); therefore, the salinity injection causes the carbon reservoir originally stored in the deep sea to be transported toward the direction of weakened stratification (i.e., toward the upper ocean), entering the PMOC-controlled cell. As the deep PMOC transports southward, the positive DIC anomalies enter the Southern Ocean and, along with the thermohaline circulation, then enter the North Atlantic. In the Shallow Convection Experiment, the North Atlantic experiences slight carbon loss, with a mechanism similar to the Pacific, namely that the contraction of AMOC is accompanied by the strengthened ocean interior stratification, weakening the upward transport capacity of DIC.

Processes in the Southern Ocean are more complex. In the previous subsection, we have already observed that the surface salinity changes in the Southern Ocean are small, with temperature rising, and the surface temperature-salinity changes are unfavorable for the observed enhancement of convection. Therefore, its dynamical mechanism mainly comes from the ocean interior. Taking the Deep Convection Experiment as an example, the salinity structure shows that the deep Southern

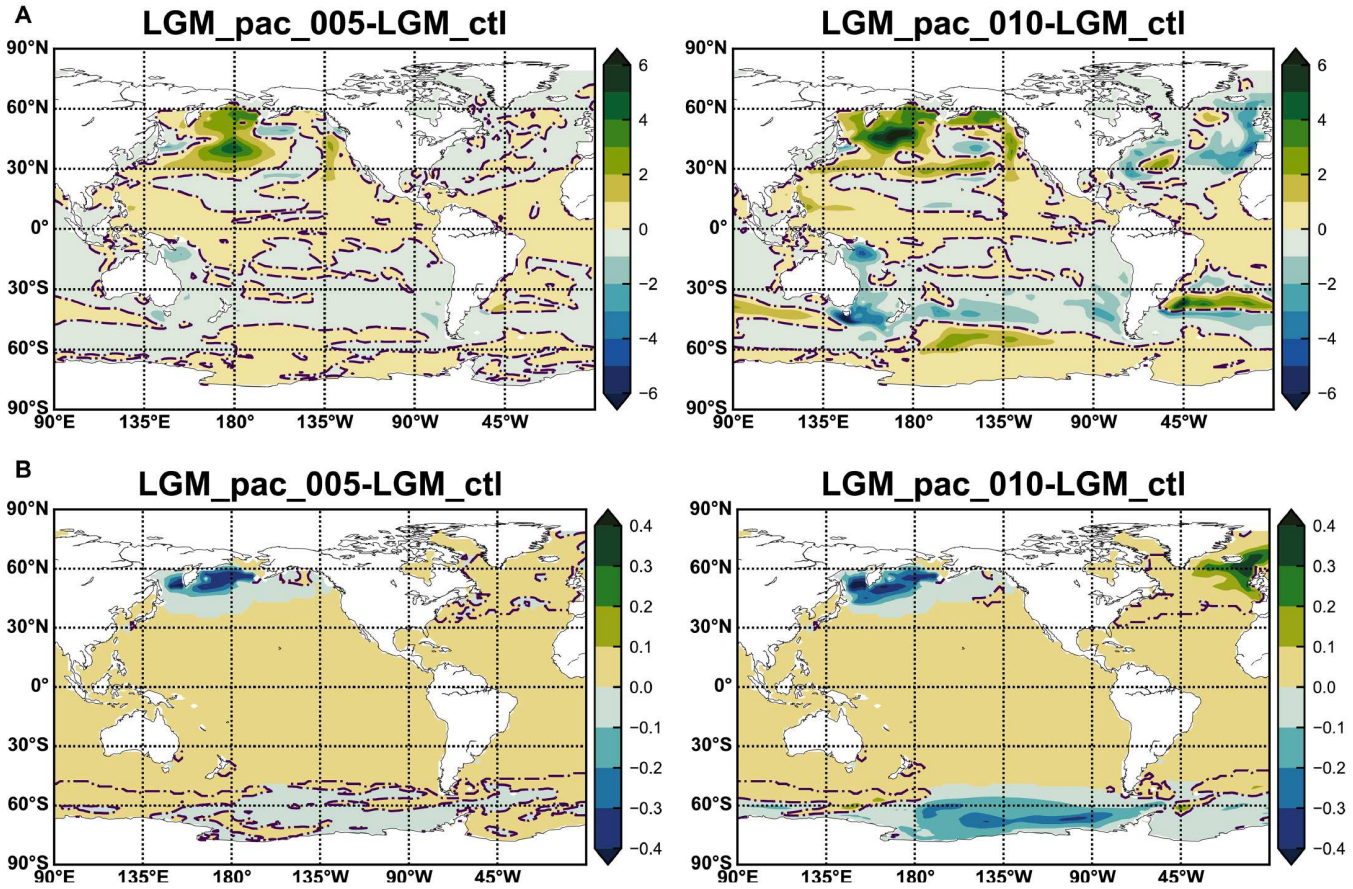


Fig. 6 The sea surface physical field anomalies in the sensitivity experiments compared to the Glacial Control Experiment. (A) Sea surface temperature (SST, units: °C); (B) Sea ice concentration (SIC, dimensionless).

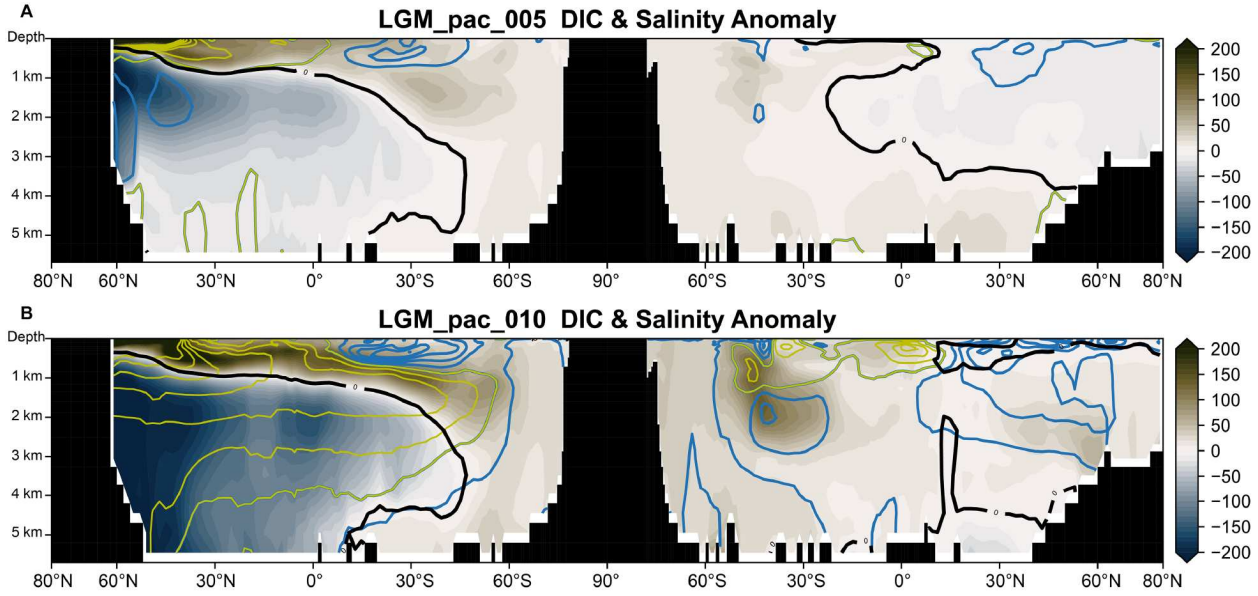


Fig. 7 Response of ocean carbon storage to the glacial Pacific halocline perturbation, shown as differences between sensitivity experiments and the Glacial Experiment. The shading in the figure represents the basin-scale zonally averaged dissolved inorganic carbon (DIC, units: mmol m^{-3}), and contours represent salinity (units: psu). From left to right, the panels are for the Pacific, Southern Ocean, and Atlantic, respectively.

Ocean also experiences an overall salinity decline, and the decline in deep salinity is greater than in the upper ocean, thereby weakening the stratification in the Southern Ocean interior, making a stronger SOMOC possible. SOMOC plays an important

role in the ocean carbon cycle. SOMOC recirculates carbon from the end of thermohaline circulation (i.e., the deep North Pacific) back to the Southern Ocean surface (Yu et al., 2020), serving as the main pathway for the re-ventilation of deep ocean water masses and "old carbon". Therefore, enhanced SOMOC will increase the ratio of carbon transported outward from the North Pacific interior. Meanwhile, the expansion of PMOC also brings more high-salinity, carbon-rich intermediate waters to the Southern Ocean, further enhancing carbon storage in the Southern Ocean.

3.4 Surface carbon cycle process response

Changes in the North Pacific halocline structure affect both the deep and surface ocean carbon cycles and have significant impacts on regional carbon source—sink patterns. In this subsection, we discuss the controlling role of the glacial North Pacific halocline on surface carbon cycle processes. To this end, we start with three variables in the sensitivity experiments: surface nutrients, export production, and surface carbon exchange flux.

The response of the surface carbon cycle is first manifested in changes in surface nutrients (Fig. 8). Similar to DIC, the distribution and transfer of nutrients in the ocean are also controlled by physical processes and regional biogeochemical processes. In both convection experiments, the response of thermohaline circulation is represented by PMOC formation and SOMOC enhancement. Since nutrient concentrations in the ocean interior are higher than those at the surface, enhanced ocean circulation facilitates the transport of nutrients from the ocean interior to the surface, which is prominently reflected in the simulated differences in surface dissolved inorganic nitrogen (DIN) and surface dissolved iron (DFe) fields. In the Shallow Convection Experiment, as the salinity flux increases and circulation strength enhances, the surface DIN and DFe concentrations in the North Pacific and Southern Ocean generally rise, yet the surface concentration changes in the North Atlantic are minimal. In the Deep Convection Experiment, the further intensified circulation continues to elevate surface nutrient concentrations, with positive nutrient anomalies gradually expanding from high latitudes to low latitudes. This phenomenon has two reasons: first, the enhancement of Pacific upwelling brings more nutrients; second, in the Deep Convection Experiment, unused nutrients in high-latitude oceans increase, thereby migrating with surface currents to other ocean regions.

Changes in nutrients, especially in surface DFe, further drive the evolution of surface primary productivity. Primary production processes (mainly photosynthesis) are the main pathway for the marine biological carbon pump to sequester CO_2, while the efficiency of primary productivity in nutrient utilization is largely limited by iron (Moore et al., 2013). Here, we take the export production—the flux of particulate organic carbon (POC) leaving the upper 100 m of the ocean, as an example to reflect the response of productivity to changes in nutrients (Fig. 9). In the sensitivity experiments, the additional iron supply brought by the enhancement of PMOC and SOMOC increases the export production in the North Pacific and Southern Ocean. In the equatorial eastern Pacific, the changes in export production in

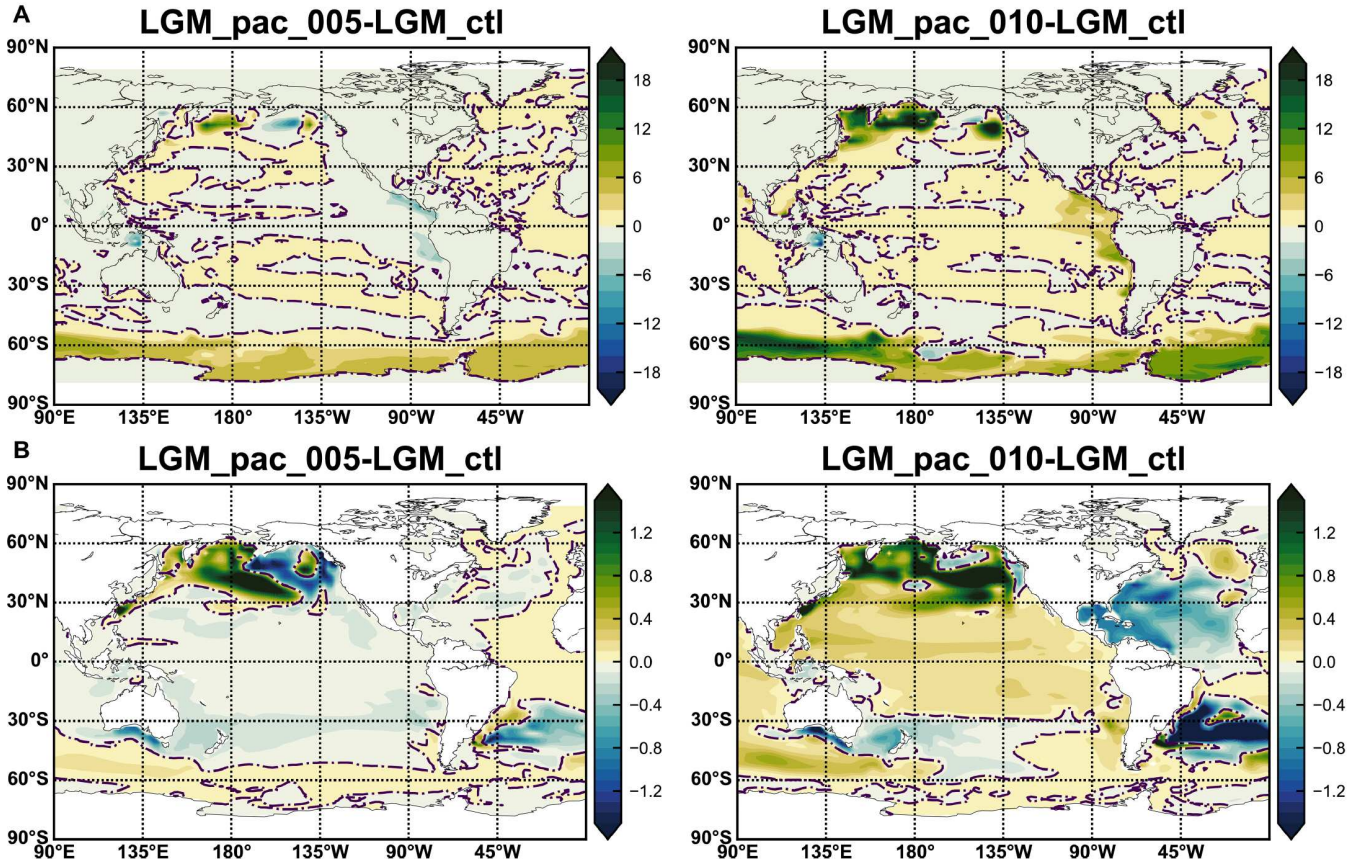


Fig. 8 Surface nutrient anomalies in the sensitivity experiments compared to the Glacial Control Experiment (units: mmol m^{-3} for DIN, μmol m^{-3} for DFe).

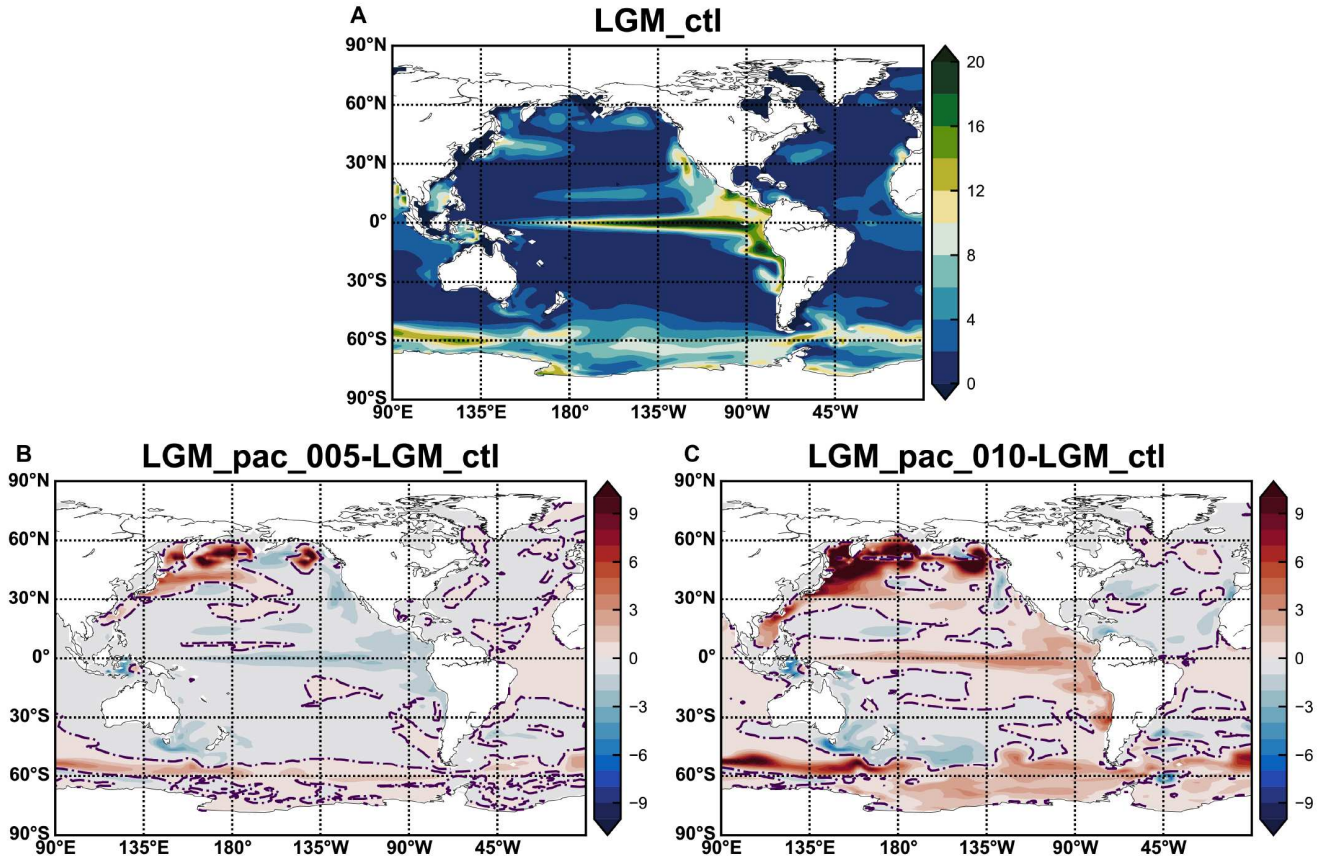


Fig. 9 Export production anomalies in the sensitivity experiments compared to the Glacial Control Experiment (units: mol C m^{-2} day^{-1}).

the Shallow Convection Experiment are not significant; whereas in the Deep Convection Experiment, due to the further strengthening of Pacific upwelling, the export production in the equatorial eastern Pacific also increases accordingly. However, it should be noted that an increase in export production does not necessarily mean that the regional ocean will store more carbon, because while the upwelling supplies nutrients from the ocean interior, it also transports DIC upward from the ocean interior, thereby causing CO_2 release at the sea surface through physical processes. The biological carbon sequestration process determines how many nutrients are consumed at the ocean surface, thereby offsetting or even exceeding the CO_2 brought by ocean physical processes. In the convection experiments, the growth of surface nutrients in the equilibrium state implies that the biological pump did not completely offset the upwelling of CO_2, thus, ultimately, the ocean releases CO_2, causing atmospheric CO_2 to rise instead.

Physical processes and biogeochemical feedbacks further lead to responses in the sea surface carbon source—sink patterns. Sea surface ΔpCO_2 represents the difference in CO_2 partial pressure between the sea surface and atmosphere (Fig. 10), serving as an important indicator of carbon source—sink patterns: in regions where $\Delta pCO_2>0$, the ocean tends to release CO_2; conversely, when $\Delta pCO_2<0$, the ocean absorbs carbon. In the Glacial Experiment, the North Atlantic and North Pacific are carbon sinks ($\Delta pCO_2<0$), while the Southern Ocean acts as a carbon source ($\Delta pCO_2>0$). The carbon sink pattern in the high-latitude Northern Hemisphere is primarily controlled by significant cooling (Yu et al., 2019), whereas the carbon source pattern in the Southern Ocean is controlled by Southern Ocean temperature, sea ice, and circulation patterns (Marzocchi and Jansen, 2019; Du et al., 2022). In the sensitivity experiments, the carbon sink in most areas of the North Pacific (except 45°N–50°N, see analysis below) and the carbon source in the Southern Ocean are both enhanced. An obvious question is: when both PMOC and SOMOC strengthen in the sensitivity experiments, why do the carbon source-sink feedbacks at the northern and southern ends of the Pacific exhibit stark differences?

Further analysis indicates that the differences in ΔpCO_2 are primarily controlled by re-equilibration of the marine inorganic carbon system. The factors influencing ΔpCO_2 are mainly sea surface DIC (Fig. 7), alkalinity (ALK, Fig. 11A), SST (Fig. 6A), and SSS (Fig. 3). Under otherwise unchanged conditions, an increase in ALK enhances CO_2 solubility, thereby lowering sea surface pCO_2; whereas increases in DIC, SST, and SSS elevate sea surface pCO_2. In the Southern Ocean, the convection experiments generally show significant SST rise, DIC increase, overall unchanged SSS, and basically unchanged ALK (Shallow Convection Experiment) or slight increase (Deep Convection Experiment). Thus, when SST and DIC jointly elevate sea surface pCO_2, although ALK may rise slightly (as in Deep Convection Experiment), it does not reverse the trend in ΔpCO_2. In the North Pacific, there is a significant rise in ALK in the convection experiments, thereby substantially lowering sea surface pCO_2, exceeding the pCO_2 changes caused by the SSS rise induced by sea surface salinity flux, the DIC rise induced by convection, and

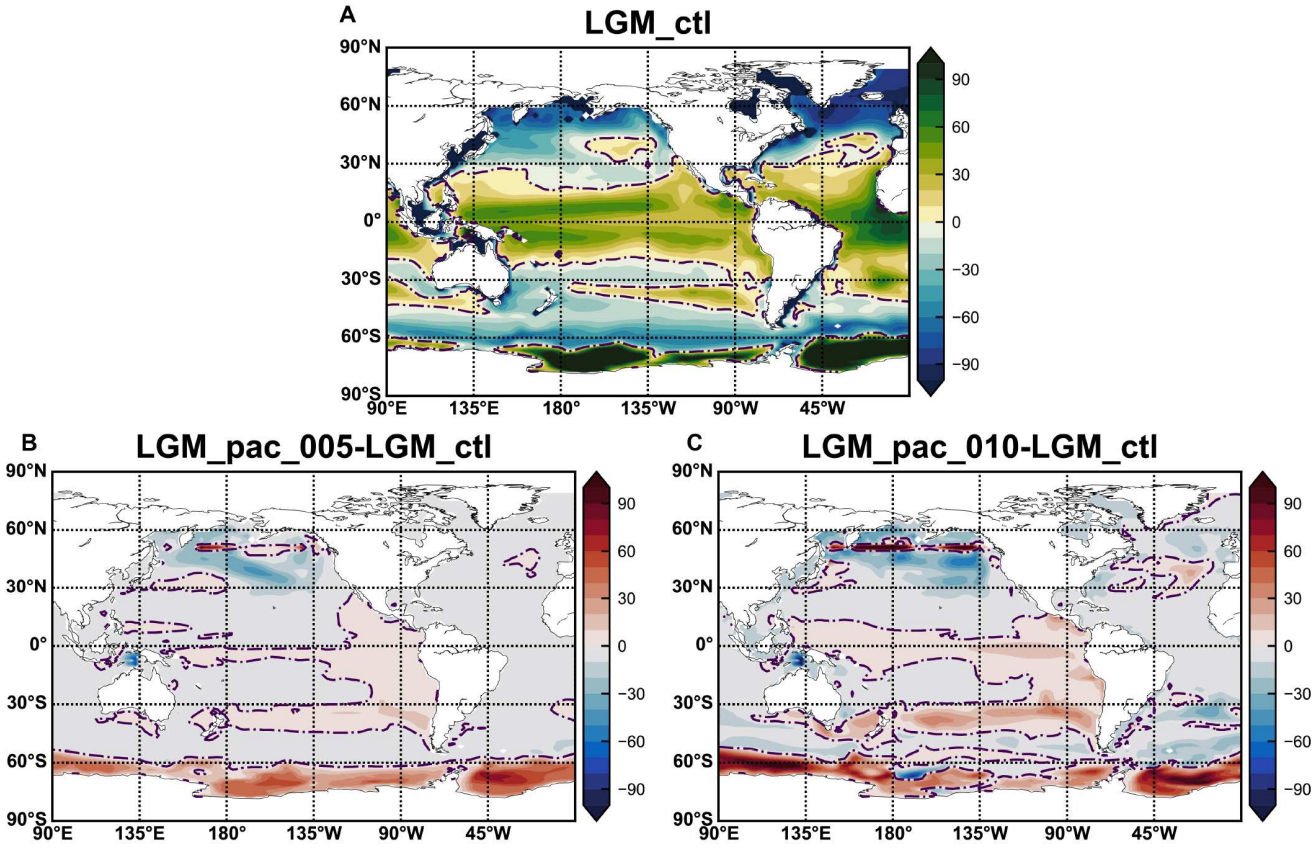


Fig. 10 Simulated sea surface ΔpCO_2 in the Glacial Experiment and the anomalies in the sensitivity experiments, units: ppm.

the North Pacific SST rise induced by PMOC, leading to enhanced carbon sink in most areas of the North Pacific.

It is noteworthy that in both convection experiments, the North Pacific region between 45°N and 50°N exhibits progressively enhanced positive ΔpCO_2 anomalies (Fig. 10), meaning its carbon sink role is weakened, with its distribution corresponding exactly to the regions of maximum positive SSS anomalies (Fig. 3) and most significant MLD deepening (Fig. 4). In the Deep Convection Experiment, both DIC and ALK concentrations in this region rise. However, only in these regions with the greatest MLD deepening is the rise in ALK concentration less than the rise in DIC concentration (Fig. 11B shows ΔALK-ΔDIC in the sensitivity experiments). Since ALK and DIC have opposite effects on sea surface ΔpCO_2, but with roughly equivalent coefficients (Menviel et al., 2012). If the ALK increase is less than that of DIC, it will cause the pCO_2 increase from DIC to be greater than the pCO_2 decrease from ALK, further compounded by the effects of SSS and SST growth (pCO_2 rise), thus the carbon sink in these areas is weakened. This unique band of ΔpCO_2 pattern further indicates that the carbon cycle processes in the North Pacific are closely related to the specific configuration of regional convection, and are controlled by both ALK and DIC in the marine inorganic carbon system (Du et al., 2025).

Finally, on the basin scale, the differences in sea surface carbon flux relative to the Glacial Experiment are shown in Fig. 12. In Fig. 12, positive values represent weakened ocean carbon absorption or enhanced carbon release, while negative values represent enhanced ocean carbon absorption or weakened release. As mentioned earlier, the regional source—sink patterns in the North Pacific and Southern Ocean are enhanced. The carbon sink effect in the North Atlantic is somewhat weakened in both experiments, mainly because in the experiments, AMOC weakens under the seesaw effect of the Atlantic—Pacific MOC. As the primary carbon sink process in the North Atlantic, the reduced AMOC weakens the North Atlantic's carbon sink effect.

3.5 Implications for glacial carbon cycles

Paleoceanographic evidence and previous numerical simulation results (Buchanan et al., 2016; Jacobel et al., 2017; Du and Tian, 2025) all point to a larger Pacific carbon reservoir, thereby contributing to the glacial atmospheric pCO_2 decline. However, research on the glacial Pacific carbon cycle dynamics still has fundamental questions that remain unresolved, with debates mainly focusing on two aspects. First is the debate over the strength of the glacial Pacific deep circulation. The classic hypothesis posits that the glacial Pacific, similar to the Atlantic, had more stagnant circulation, thereby increasing the respired carbon reservoir (Skinner et al., 2015; Du et al., 2018); however, some water mass indicators (such as Nd isotopes) propose a hypothesis of active circulation (Hu and Piotrowski, 2018). Recent numerical simulation results (Du and Tian, 2025) to some extent demonstrate that even with more active circulation, the deep Pacific carbon reservoir can still store more carbon through enhanced air-sea disequilibrium.

Surface processes in the glacial North Pacific (Gray et al., 2018; Rae et al., 2020) are even less well understood. It is generally believed that since the late Pliocene, the North Pacific

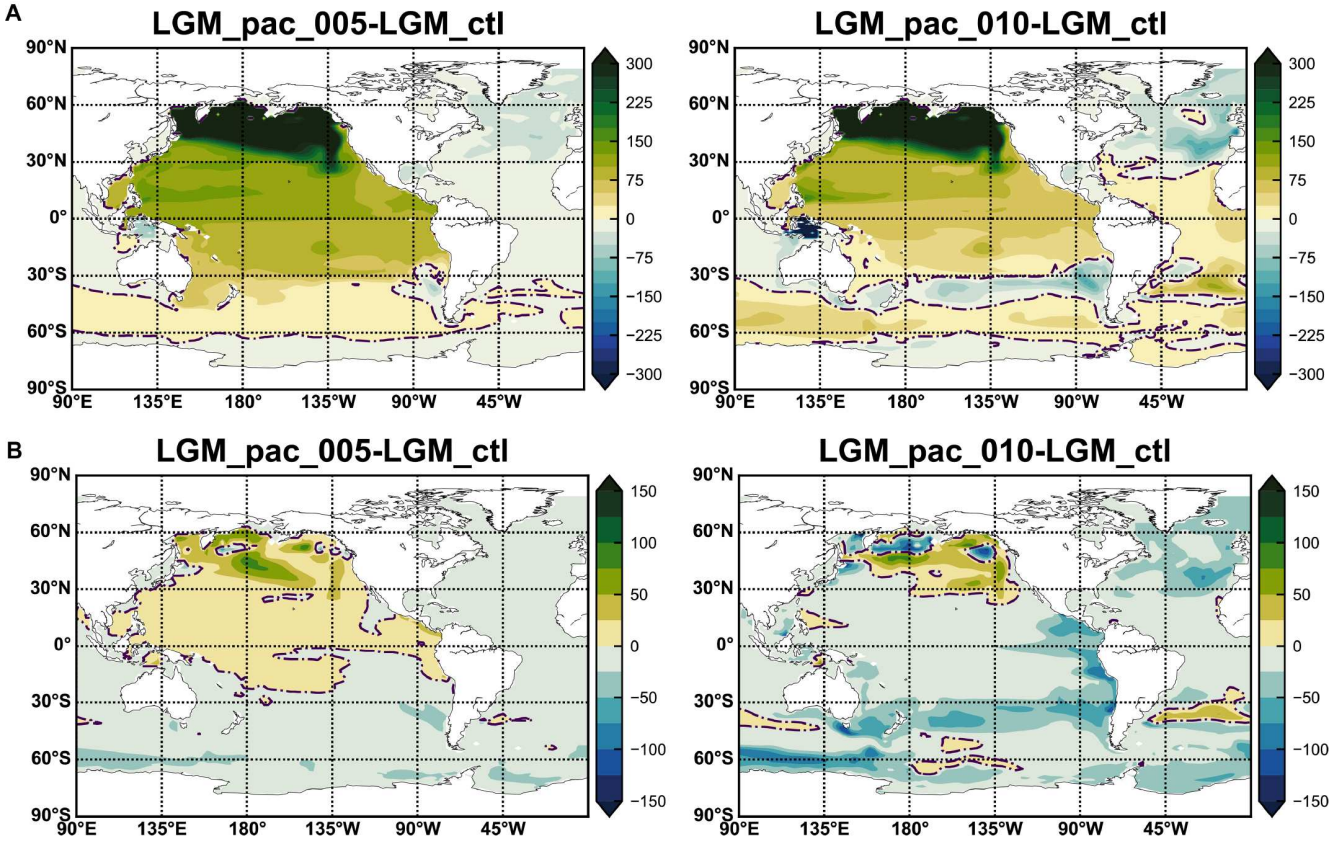


Fig. 11 Surface alkalinity (ALK) anomalies in the sensitivity experiments compared to the Glacial Control Experiment (units: mmol m^{-3}) and the ΔALK-ΔDIC differences.

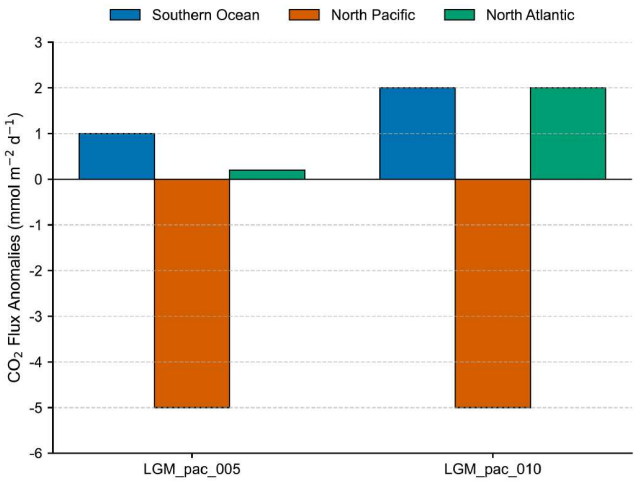


Fig. 12 Basin-scale air-sea CO_2 flux anomalies in the sensitivity experiments relative to the Glacial Experiment (units: mol C m^{-2} yr^{-1}).

halocline formed and strengthened, almost permanently isolating the deep North Pacific (Haug et al., 1999), making it the largest marine respired carbon reservoir. Recently, the work published by Rae et al. (Rae et al., 2020) compiled a series of paleoceanographic proxies and results from intermediate-complexity numerical simulations, suggesting that the glacial North Pacific was similar to modern North Atlantic, namely, with deep water formation, higher biological pump efficiency, and lower surface nutrients. The work previously published by Gray et al. (Gray et al., 2018) shows that during the LGM, the North Pacific surface carbon flux may have been small (ΔpCO_2 approaching 0), which indirectly corroborates this view. However, we find in our model that, whether in the Deep Convection or Shallow Convection Experiments, perturbation of the halocline largely causes leakage of the deep carbon reservoir (Fig. 7), allowing more CO_2 to escape from the Southern Ocean (Fig. 12), increasing atmospheric CO_2 (Table 1). This process has yielded consistent results in different numerical models under glacial or interglacial conditions (Menviel et al., 2012, 2014), which is thus not a model-dependent feature.

Our experiments indicate that deep convection in the North Pacific (as in LGM_pac_010) disrupts the stability of the deep carbon reservoir, leading to ocean carbon release. However, geological records clearly indicate that the Pacific carbon reservoir expanded during glacials (Jacobel et al., 2017). Therefore, at the current stage, there is an urgent need to reconcile the contradictions between geological records and model results. Moreover, there are even opposite views between records indicating surface ventilation (Rae et al., 2020) and deep-sea carbon storage (Jacobel et al., 2017). Based on existing results, we believe that coexistence of glacial Pacific convection and enhanced carbon storage can depend on two conditions, namely a specific mode of glacial PMOC and further clarification of the role of the Southern Ocean. In our experiments, we used a minimum of 0.05 Sv of freshwater extraction flux; but in realistic geological history, its total value may be further less than the 0.05 Sv in the Shallow Convection Experiment, thereby making the convection process in the North Pacific highly self-limiting.

Thus, the physical picture of glacial circulation would be the dominant AABW circulation expanding from the Southern Ocean to the North Atlantic and North Pacific; while the intermediate water masses formed in the North Pacific and Atlantic would lead to the existence of two carbon sinks in the Northern Hemisphere similar to the modern North Atlantic, with their strength possibly controlled by temperature (Takahashi et al., 2009). This picture can, to some extent, bridge the debates between model results and records, and maintain consistency with North Atlantic research on basin scales. On the other hand, when PMOC is active, the glacial Southern Ocean on the Pacific side may produce faster meridional circulation; at the same time, the drastic cooling of the Southern Ocean surface leads to increased sea ice, which, together with circulation speed, elevates disequilibrium pump efficiency, thereby offsetting or even reversing the adverse effects of PMOC enhancement on carbon storage (Du and Tian, 2025). In addition, the salinity contributed by sea ice increases the salinity of AABW (Marzocchi and Jansen, 2019), making it more difficult for PMOC to disrupt the bottom stratification and release the deep carbon reservoir.

Our experiments further demonstrate the complex response processes of the air-sea carbon cycle when climate states change, which we categorize into two categories of mutually coupled mechanisms:

(1) Climate-carbon cycle equilibrium determined by boundary conditions. In climate simulations, the background atmospheric pCO_2 is typically generated based on geological records and serves as an important boundary condition for the simulation, enabling climate system models to output specific atmospheric and oceanic physical fields. Carbon cycle simulations then use these physical conditions as a benchmark (especially temperature) to reveal the ocean carbon cycle modes under specific boundary conditions; these boundary conditions also determine the distribution of carbon reservoirs between the atmosphere and ocean from a 'chemical equilibrium' perspective. For example, in Table 1, although the total inventory changes in the Pacific and Atlantic in the Deep Convection Experiment are both on the order of 100 GtC, under the premise of minimal changes in average sea temperature, the final atmospheric pCO_2 change is only around 20 ppm. This phenomenon confirms that on tectonic and long orbital timescales, average ocean temperature conditions actually determine the strength of the ocean solubility pump itself (Kump and Arthur, 1999).

(2) Basin-scale carbon storage rebalancing triggered by physical conditions. Changes in glacial Pacific ventilation states or PMOC are essentially carbon reservoir transfers between basins, between surface and deep ocean, triggered by physical conditions (Du et al., 2025). During the transfer process, carbon chemical equilibrium is jointly modulated by inorganic processes (solubility pump, carbonate chemistry) (Anderson et al., 2019) and organic processes (photosynthesis and respiration) (Khatiwala et al., 2019), so that this carbon transfer between basins ultimately causes the ocean carbon reservoir to deviate from its original equilibrium state, altering atmospheric CO_2 through carbon balance responses with the atmosphere (or land), thereby combining with the aforementioned mechanism to initiate a new climate-carbon cycle rebalancing process.

4. Conclusions and outlook

This study, based on the state-of-the-art ocean-biogeochemical model MITgcm-REcoM2, conducted a set of three experiments under idealized LGM glacial conditions, revealing the interactions between the North Pacific halocline and the deep carbon reservoir under glacial conditions. Our results indicate that, at freshwater extraction intensities of 0.05 and 0.1 Sv, weakening or removing the North Pacific halocline directly results in the generation of PMOC and the strengthening of SOMOC. Enhanced circulation, transport, and diapycnal mixing are unfavorable for the enrichment and maintenance of the glacial Pacific carbon reservoir. In this scenario, surface ocean carbon cycle processes are simultaneously regulated by physical conditions (SST, SSS) and carbon chemical equilibrium (ALK, DIC), with inorganic carbon chemical equilibrium processes dominating in the North Pacific. Our results indicate that if intermediate-depth convection exists in the glacial North Pacific, it may make the North Pacific a weak carbon sink; however, its specific climate dynamics driving mechanisms, as well as its interactions with North Atlantic deep water formation, still require more work to reveal.

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Compliance and ethics

The authors declare no conflict of interest.

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